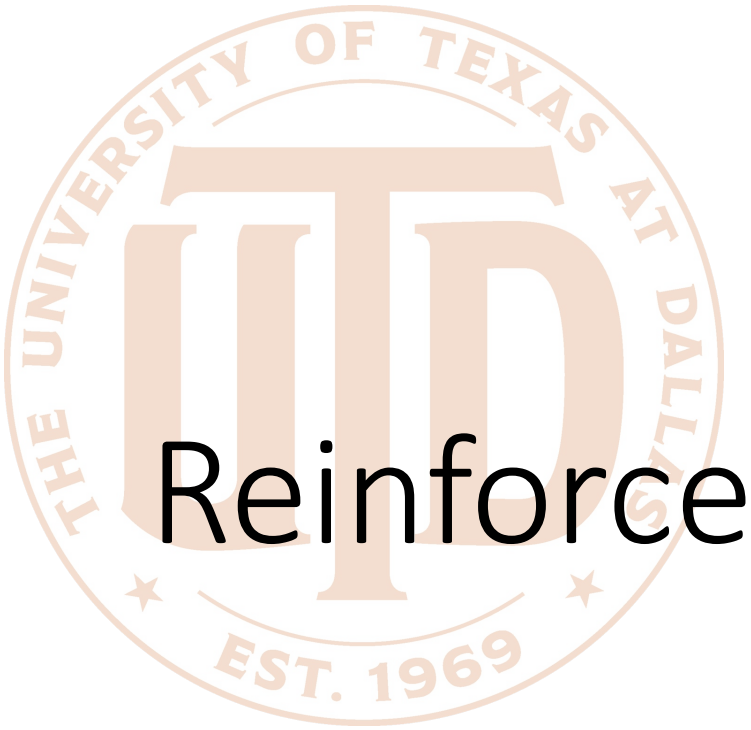


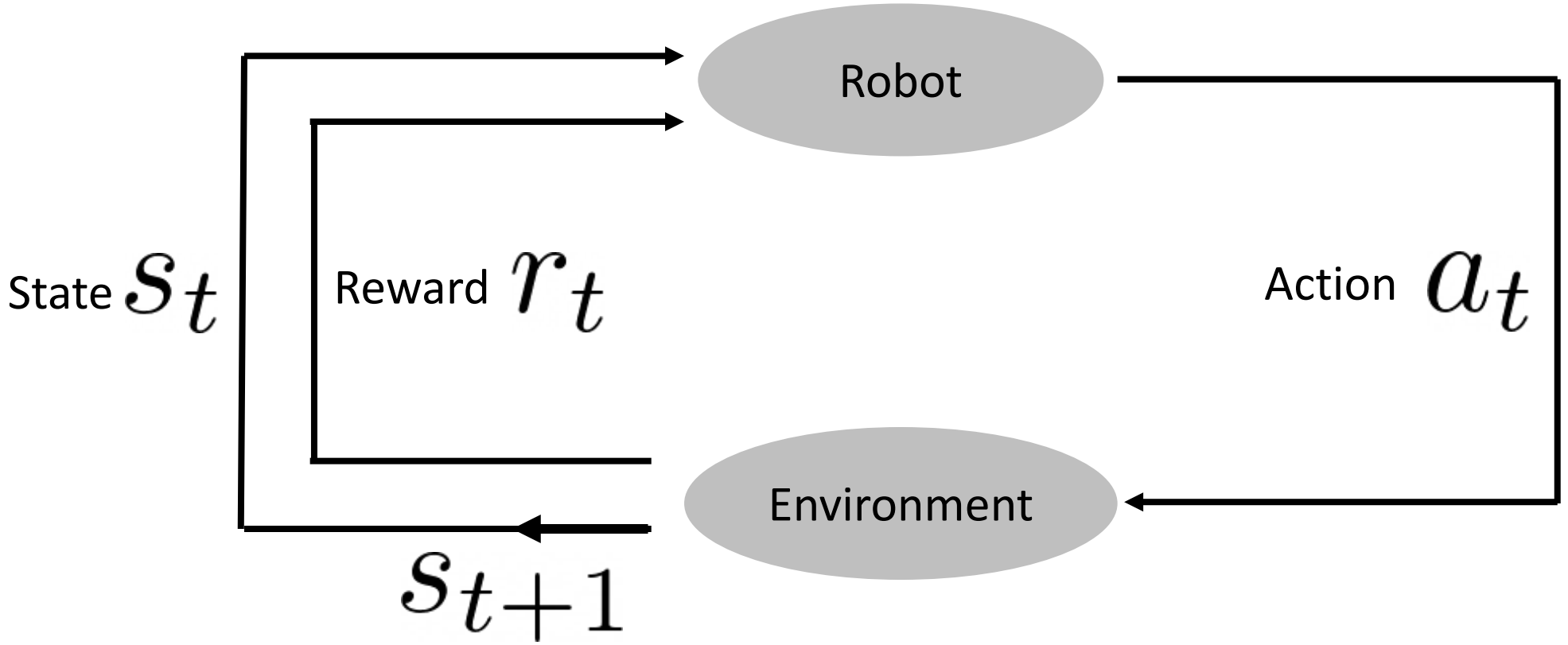
Reinforcement Learning: Concepts

CS 6341 Robotics

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Reinforcement Learning



Reinforcement Learning:

Imitation Learning: $a_t = \pi(s_t)$

The RL Problem

- The goal of RL is to select a policy which maximizes expected return when the agent acts according to it
- Probability distribution over trajectories

$$\tau = (s_0, a_0, r_0, s_1, a_1, r_1, \ldots)$$

$$P(\tau|\pi) = \rho_0(s_0) \prod_{t=0}^{T-1} P(s_{t+1}|s_t, a_t)\pi(a_t|s_t)$$

Transition model (no need in model-free RL)

$$p(s' \mid s, a)$$

- Expected return

$$J(\pi) = \int_{\tau} P(\tau|\pi)R(\tau) = \mathop{\mathrm{E}}_{\tau\sim\pi}[R(\tau)]$$

Sample trajectories

$$R(\tau) = \sum_{t=0}^{T} r_t \qquad R(\tau) = \sum_{t=0}^{\infty} \gamma^t r_t$$

$$\gamma \in (0, 1)$$

- The central optimization problem

$$\pi^* = \arg\max_{\pi} J(\pi)$$

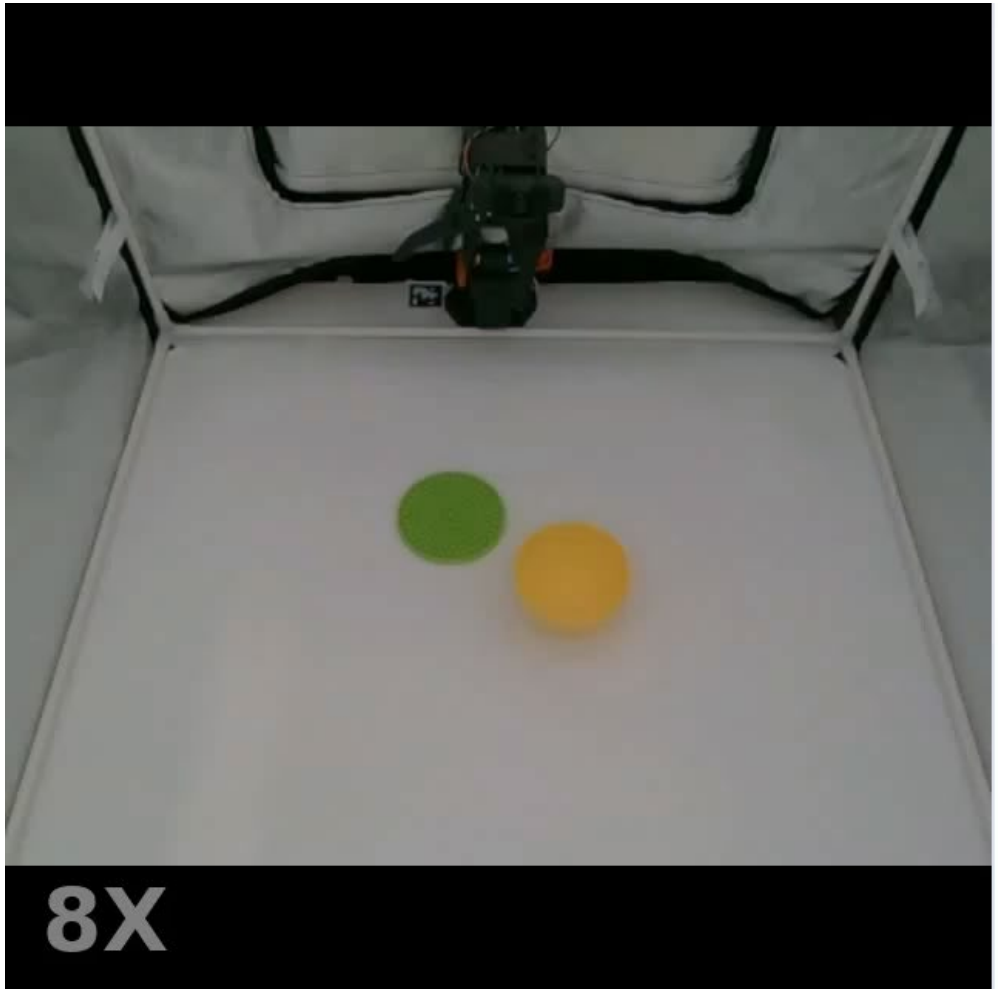
Optimal policy

In practice

$$\pi_\theta^* = \arg\max_{\theta} J(\pi_\theta)$$

Learn the parameters of the policy

SO101 Example





Value Functions

- Value of a state or a state-action pair
  - The expected return if you start in that state or state-action pair, and then act according to a particular policy forever after
- On-policy Value Function $V^{\pi}(s) = \underset{\tau \sim \pi}{\mathrm{E}} \left[R(\tau) \,|s_0 = s \right]$
- On-policy Action-Value Function $Q^{\pi}(s,a) = \underset{\tau \sim \pi}{\mathrm{E}} \left[R(\tau) \,|s_0 = s, a_0 = a \right]$
- Optimal Value Function $V^{*}(s) = \max_{\pi} \underset{\tau \sim \pi}{\mathrm{E}} \left[R(\tau) \,|s_0 = s \right]$
- Optimal Action-Value Function $Q^{*}(s,a) = \max_{\pi} \underset{\tau \sim \pi}{\mathrm{E}} \left[R(\tau) \,|s_0 = s, a_0 = a \right]$

Value Functions

- Connection

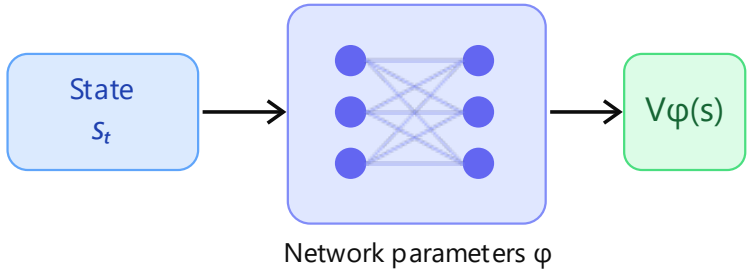
$$V^{\pi}(s) = \mathop{\mathrm{E}}_{a \sim \pi}[Q^{\pi}(s,a)] \qquad V^{*}(s) = \max_{a} Q^{*}(s,a)$$

- The optimal policy in S will select whichever action maximizes the expected return starting in S

$$a^{*}(s) = \arg\max_{a} Q^{*}(s,a)$$

Parametrized Value Functions

- On-policy Value Function $V^{\pi}(s) = \underset{\tau \sim \pi}{\mathrm{E}} \left[R(\tau) \,|s_0 = s \right]$
- Parameterization (a network) $V_{\phi}(s)$



- Learning the value function
  - Sample trajectories $\tau = (s_0, a_0, r_0, s_1, a_1, r_1, \ldots)$
  - For each trajectory $G_t = \sum_{k=t}^{T} \gamma^{k-t} r_k$
  - Supervised learning $L(\phi) = \frac{1}{N} \sum_{t} \left(V_{\phi}(s_t) - G_t \right)^2$

Bellman Equations

- The value of your starting point is the reward you expect to get from being there, plus the value of wherever you land next

On-policy

$$V^{\pi}(s) = \mathop{\mathrm{E}}_{\substack{a\sim\pi \\ s'\sim P}}\left[r(s,a) + \gamma V^{\pi}(s')\right],$$

$$Q^{\pi}(s,a) = \mathop{\mathrm{E}}_{s'\sim P}\left[r(s,a) + \gamma \mathop{\mathrm{E}}_{a'\sim\pi}\left[Q^{\pi}(s',a')\right]\right]$$

Q Estimation

$$\hat{Q}(s_t, a_t) = r_t + \gamma V_{\phi}(s_{t+1})$$

Optimal policy

$$V^{*}(s) = \max_{a} \mathop{\mathrm{E}}_{s'\sim P}\left[r(s,a) + \gamma V^{*}(s')\right],$$

$$Q^{*}(s,a) = \mathop{\mathrm{E}}_{s'\sim P}\left[r(s,a) + \gamma \max_{a'} Q^{*}(s',a')\right]$$

Advantage Functions

- How much better it is to take a specific action a in state s, over randomly selecting an action according to $\pi(\cdot|s)$

$$A^{\pi}(s,a) = Q^{\pi}(s,a) - V^{\pi}(s)$$

Q value for (s, a)

V value for s: random action from the policy

$$Q^{\pi}(s,a) = \mathop{\mathrm{E}}_{\tau\sim\pi}\left[R(\tau)\,|s_0 = s, a_0 = a\right]$$

$$V^{\pi}(s) = \mathop{\mathrm{E}}_{\tau\sim\pi}\left[R(\tau)\,|s_0 = s\right]$$

Advantage Estimation

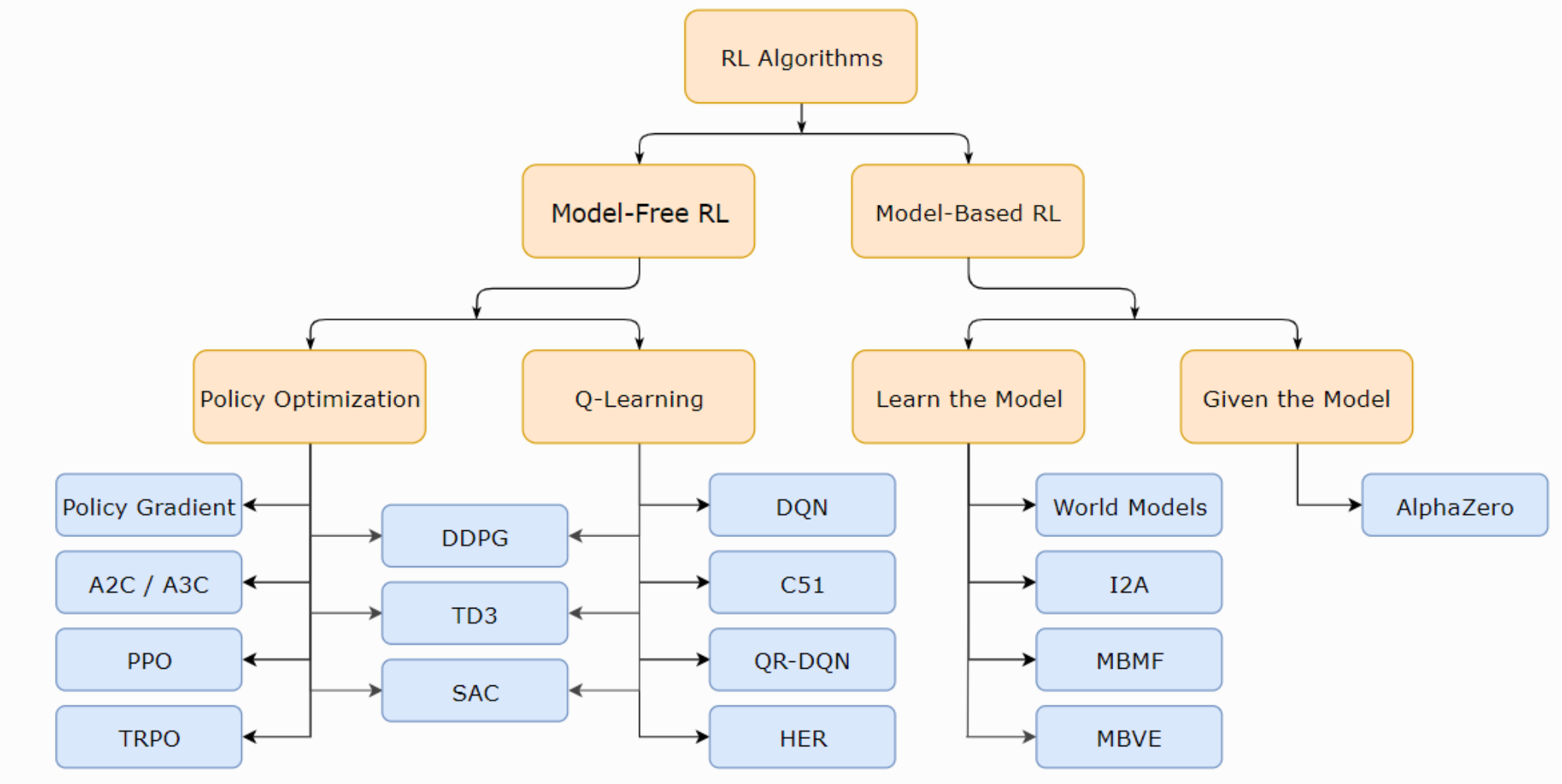
$$A^{\pi}(s,a) = Q^{\pi}(s,a) - V^{\pi}(s)$$
$$= r + V^{\pi}(s') - V^{\pi}(s)$$

Markov Decision Processes (MDPs)

- A MDP is a 5-tuple $\langle S, A, R, P, \rho_0 \rangle$

- S is the set of all valid states,
- A is the set of all valid actions,
- $R : S \times A \times S \to \mathbb{R}$ is the reward function, with $r_t = R(s_t, a_t, s_{t+1})$,
- $P : S \times A \to \mathcal{P}(S)$ is the transition probability function, with $P(s'|s, a)$ being the probability of transitioning into state s' if you start in state s and take action a,
- and ρ_0 is the starting state distribution.

A Taxonomy of RL Algorithms



Model-Free vs. Model-based RL

- Whether the agent has access to (or learns) a model of the environment

- A model is a function which predicts state transitions and reward
  - Transition model $p(s' \mid s, a)$
  - Reward model $r(s, a)$

- A model allows the agent to plan by thinking ahead

- A ground-truth model of the environment is usually not available to the agent

Further Reading

- OpenAI Spinning Up in Deep RL
  https://spinningup.openai.com/en/latest/index.html